STATE OF NEW YORK
PUBLIC SERVICE COMMISSION

Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of The Brooklyn Union Gas Company d/b/a National Grid NY for Gas Service	**Case 19-G-0309**
Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of KeySpan Gas East Corp. d/b/a Brooklyn Union of L.I. for Gas Service	**Case 19-G-0310**

Prepared Direct Testimony of:

New York City Energy Efficiency Panel

John H. Lee
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On Behalf of:

The City of New York

August 30, 2019

INTRODUCTION

Q. PLEASE STATE YOUR NAME, OCCUPATION AND BUSINESS ADDRESS.

A. Mr. Lee – My name is John H. Lee. I am the Deputy Director for Green Buildings and Energy Efficiency at the New York City Mayor's Office of Sustainability. My business address is 253 Broadway, 14th floor, New York, New York.

Ms. Kelly – My name is Elizabeth D. Kelly. I am a Senior Policy Advisor at the New York City Mayor's Office of Sustainability. My business address is 253 Broadway, 14th floor, New York, New York.

Ms. Roberton – My name is Jennifer Roberton. I am the Transportation Policy Advisor at the New York City Mayor's Office of Sustainability, or MOS. My business address is 253 Broadway, 14th floor, New York, New York.

Q. ON WHOSE BEHALF ARE YOU SUBMITTING THIS TESTIMONY?

A. We are submitting this direct testimony before the New York Public Service Commission ("PSC" or the "Commission") on behalf of the City of New York ("City").

Q. PLEASE DESCRIBE YOUR EDUCATIONAL BACKGROUND.

A. Mr. Lee – I graduated from Rice University with a Bachelor of Architecture Degree and from Harvard University with a Master of Architecture Degree. I am a Leadership in Energy and Environmental Design - Accredited Professional certified by the U.S. Green Building Council. I am a licensed architect.

Ms. Kelly – I graduated from Lake Forest College with a Bachelor of Arts Degree in Urban Studies and French and from Rutgers University with a Master of City and Regional Planning Degree.

Ms. Roberton – I graduated from the University of British Columbia with a Masters of Arts in Planning in 2016. I received an Honours Bachelor of Arts from the University of Toronto. I am an American Institute of Certified Planners Candidate Member, as certified by the American Planning Association.

Q. PLEASE DESCRIBE YOUR PROFESSIONAL EXPERIENCE AND CURRENT RESPONSIBILITIES.

A. Mr. Lee – I commenced my public service career with the New York City Department of City Planning as an Urban Designer. From there I moved to the New York City Department of Buildings as a Senior Architect in the codes development divisions. In December 2012, I joined what was then known as the New York City Mayor's Office of Long Term Planning and Sustainability and now known as the Office of Sustainability. I have had positions of increasing responsibility in this Office. In my current capacity, I lead the City's policy and legislative efforts driving the building environment to energy efficiency standards necessary to achieve the City's greenhouse gas reduction goals.

Ms. Kelly – I spent two years organizing community and economic development programs for the North River Commission in Chicago. Subsequently, I was employed by The Community Preservation Corporation for seven years working on sustainability issues. I held positions of increasing responsibility, including

Assistant Vice President for Sustainability Programs. My responsibilities included developing and implementing strategies to leverage private capital in support of environmental sustainability in multifamily affordable housing development projects, engaging with stakeholder and business and community partners, and providing training and technical assistance for sustainability projects. In 2019, I joined the Mayor's Office of Sustainability, where my responsibilities include developing public policies and related programs to reduce greenhouse gas emissions from New York City's public and private buildings and other structures.

Ms. Roberton – I commenced my career in transportation in 2010 as the Education Coordinator at Bikechain, a non-profit mobility access organization based in Toronto, Ontario, Canada. I began working in public service in 2016 as a Program Specialist at the Environment and Energy Division at the City of Toronto. In November 2017, I moved to New York to join MOS as the Transportation Policy Advisor. I lead the Office's transportation portfolio, including the development of a citywide network of electric vehicle, or EV, chargers through policy, infrastructure projects and legislative efforts. A key part of this initiative is the implementation of a $10 million project to install EV fast chargers in every borough, which I manage in partnership with New York City's Department of Transportation, or NYCDOT. I also manage the City's comments to the Public Service Commission on matters regarding vehicle electrification.

Q. HAVE YOU EVER TESTIFIED BEFORE ANY AGENCY OR COURT OR IN ANY OTHER VENUE ON THE TOPICS YOU ADDRESS IN THIS TESTIMONY?

A. Mr. Lee – I have submitted testimony to the Public Service Commission in prior The Brooklyn Union Gas Company d/b/a National Grid NY ("KEDNY") and KeySpan Gas East Corp. d/b/a National Grid ("KEDLI") rate cases. I also recently testified on behalf of the City in the Consolidated Edison Company of New York, Inc. ("Con Edison") 2019 electric and gas rate cases. In addition, I have testified before New York City Council in support of legislation related to energy efficiency in buildings and resiliency. I also frequently give talks at conferences on energy use information, such as benchmarking and energy audits, and energy efficiency issues, revisions to energy codes, and building retrofit measures, as well as NYC Clean Heat and the NYC Retrofit Accelerator.

Ms. Kelly – I have testified before New York City Council in support of legislation related to energy efficiency in buildings and clean energy financing. I also have presented at conferences on topics related to energy. I also testified in the 2019 Con Edison rate cases.

Ms. Roberton – I recently testified on behalf of the City in the 2019 Con Edison rate cases. I also frequently speak at conferences on sustainable transportation, including vehicle electrification, freight efficiency, and mode shift away from private vehicle use, as well as on the City's fast charging program and curbside projects conducted in partnership with Con Edison.

PURPOSE OF TESTIMONY

Q. WHAT IS THE PURPOSE OF YOUR TESTIMONY?

A. The purpose of our testimony is to discuss the City's public policies related to energy efficiency and their interaction with the Companies' various proposals set forth in the direct testimony of the Future of Heat Panel.

Q. PLEASE SUMMARIZE YOUR TESTIMONY AND RECOMMENDATIONS.

A. The City and State of New York have corresponding policies that encourage increasing reliance on energy efficiency to reduce greenhouse gas emissions and the need for fossil-fueled power plants. In this testimony, we recommend changing the structure and scope of energy efficiency programs to target more comprehensive building retrofits, to help to achieve the energy efficiency targets established by the Public Service Commission in its December 13, 2018 Order in Case 18-M-0084, or Energy Efficiency Order. We also make recommendations on the intersection between the Companies' energy efficiency programs and the City's Retrofit Accelerator. We further provide input on the Companies' proposed expanded Demand Response Demonstration Project, as well as their proposed Power-to-Gas ("P2G") program. We then make recommendations on the Companies' efforts to electrify their fleets. Finally, we express concerns about the size and structure of the Companies' shareholder incentive proposals. Any incentives should be linked to superior efforts by the utilities, not business-as-

usual conduct or circumstances outside of the utilities' control, and the Companies should not receive multiple/duplicative incentives for the same outcomes.

Q. ARE YOU SPONSORING ANY EXHIBITS WITH YOUR TESTIMONY?

A Yes. We are sponsoring Exhibit__(NYCEEP-1), which is a compilation of discovery responses we relied upon in preparing our testimony, some of which we expressly reference herein.

ENERGY EFFICIENCY INCENTIVES FOR RETROFITS

Q. HAVE YOU REVIEWED THE COMPANIES' PROPOSED ENERGY EFFICIENCY PROGRAMS, BUDGETS, AND TARGETS?

A. Yes.

Q. WHAT BUDGETS AND TARGETS DO THE COMPANIES PROPOSE?

A. The Companies propose to maintain their current annual energy efficiency transition implementation plan ("ETIP") budgets through 2020. The Companies also propose using previously-adopted budget amounts for 2020, and the presumptive targets and budgets for 2021-2025, set forth in the Energy Efficiency Order, to determine their proposed budget amounts to achieve incremental energy efficiency savings for the Rate Year and Data Years.

Q. HOW WERE THE BUDGETS AND TARGETS FOR INCREMENTAL ENERGY EFFICIENCY SAVINGS ESTABLISHED?

A. In the Energy Efficiency Order, the Commission stated that the budgets for increased utility energy efficiency efforts were based on the lesser of Commission-authorized or current actual run rates for each utility's existing portfolio. In this

context, run rates for non-low-to-moderate income ("LMI") customers represent the ETIP portfolio level expended funds divided by the portfolio level acquired savings through the fourth quarter of 2017.

Q. ARE THE PROPOSED INCREMENTAL ENERGY EFFICIENCY TARGETS APPROPRIATE?

A. No. The use of historic run rates to establish the Companies' incremental MMBtu savings does not accurately reflect the Companies' actual, recent unit costs to achieve energy efficiency savings. For example, as shown in the Companies' response to City Information Request ("IR") 123, in 2018, KEDNY had an average unit cost of $1.64 per therm of energy efficiency savings, while KEDLI had an average unit cost of $1.61. Over the last two years, the average per-therm unit costs for KEDNY and KEDLI are $1.86 and $1.87, respectively.

In comparison, the Energy Efficiency Order set KEDNY incremental non-LMI MMBtu savings target for 2019-2020 at 231,401 MMBtu, with a total budget of $6,452,620. KEDLI's incremental non-LMI MMBTu savings target for the same two-year period was set at 112,554 MMBtu, with a total budget of $2,674,824. Based on these targets and budgets, the per-therm costs for KEDNY and KEDLI are approximately $2.79 and $2.38, respectively.

Q. HOW SHOULD THE COMPANIES' INCREMENTAL TARGETS BE CALCULATED?

A. The Companies should base their incremental energy efficiency targets on actual recent unit costs. This will encourage the Companies to continue building on recent

successes in their respective energy efficiency programs by continuing to drive down per-unit costs and increasing program effectiveness.

Q. DO YOU HAVE SPECIFIC TARGETS IN MIND?

A. Yes. The Companies should calculate their incremental energy efficiency savings targets by taking their proposed incremental budgets and dividing them by their respective energy efficiency savings unit costs over the last two years. For example, KEDNY proposes a non-LMI budget of $5,972,357 for calendar year 2021. Based on an average unit cost of $1.86 per therm saved over the last two years, the target savings for that budget should be 3,210,944 therms, or approximately 321,100 MMBtu. For the same time period, KEDLI's proposed budget is $2,297,639. Based on a two-year unit cost of $1.87, KEDNY's savings target for 2021 should be 1,228,684 therms, or approximately 122,900 MMBtu.

Q. ARE YOU ADVOCATING FOR THE COMPANIES TO FOCUS PRIMARILY ON 'LEAST COST' OR 'LOW HANGING' ENERGY EFFICIENCY MEASURES?

A. No, in fact later on in our testimony, we advocate for just the opposite – the Companies should be pursuing more comprehensive energy efficiency measures that we contend will maximize energy efficiency savings. We understand, however, that by pushing the Companies to drive down per-unit costs, we may be creating an incentive for the Companies to instead focus on 'low hanging fruit,' or energy efficiency measures that come with low costs and quick paybacks. Again, this is not the intent of our proposal. To avoid this undesirable outcome, the

Commission should adopt the suite of energy efficiency improvements that we discuss below, including payment of upfront energy efficiency incentives, stacking of incentives to increase payment to customers that pursue a combination of efficiency measures at the same time, re-designing incentives to accommodate longer timescales for building owners that undertake comprehensive energy efficiency retrofits, and improving coordination with the City's Retrofit Accelerator.

Moreover, our proposal is consistent with the Energy Efficiency Order, in that we are simply adopting the Companies' actual run rates from the last two years (calendar years 2017 and 2018) and building upon the Companies' demonstrated success in driving down per-unit costs of energy efficiency gains.

Q. PLEASE CONTINUE.

A. Our proposal is also in recognition of the need for incremental energy efficiency gains (among other measures) that has been highlighted by the stalled Northeast Supply Enhancement ("NESE") pipeline. As discussed in the direct testimony of the City's Policy Panel, and as highlighted in the Companies' response to City IR-117, the Companies have not identified any incremental energy efficiency, demand response, or other load management programs that it would plan to implement in the event that the NESE project is ultimately not built. Our proposals here, including our proposal for raising the energy efficiency targets, represent several concrete steps that the Companies can take in the immediate term to improve their energy efficiency and demand response offerings. Certainly, however, we

acknowledge that much more will be needed in the coming years to meet stringent City and State requirements on efficiency, carbon reduction, renewable energy, and the like. This is, in part, why we advocate below for the creation of a collaborative or other similar process, where interested parties can address these broader concerns in a more comprehensive manner.

Q. ARE YOU AWARE OF ANY ISSUES WITH THE CURRENT DESIGN OF THE COMPANIES' ENERGY EFFICIENCY PROGRAM OFFERINGS?

A. Yes. Based on our experience working with building owners and the utilities to implement building retrofits in New York City, we have identified a number of issues with the design of the Companies' energy efficiency program offerings that may be preventing them from achieving the maximum energy efficiency savings possible. First, in many cases, it is difficult to secure the requisite upfront financing needed to install energy efficiency improvements, a problem which is exacerbated if the building owners cannot access utility incentives until after the efficiency measure has been installed. Second, and relatedly, we believe that the Companies can achieve greater savings through their energy efficiency programs by incentivizing building owners to pursue a comprehensive package of energy efficiency programs that, together, have a greater positive impact than if the building owner only pursued individual building improvements.

Q. PLEASE EXPLAIN THE ISSUE OF UPFRONT FINANCING FOR ENERGY EFFICIENCY PROJECTS.

A. Based on our experience working with building owners to implement building retrofits in New York City, and anecdotal evidence from KEDNY and KEDLI energy efficiency program participants, the upfront cost of comprehensive energy efficiency investments can be an impediment for some customers, particularly owners of low income or rent-regulated buildings**.** Some building owners may have limited cash flow or capital reserves to make significant energy efficiency upgrades and thus must seek third party financing to fund the improvements. There are a variety of circumstances that inhibit the ability of some owners to obtain financing or increase financing costs to fund efficiency projects. As a result, these owners are unable to utilize energy efficiency incentive programs and are prevented from proceeding with projects.

Q. WHAT TYPES OF ENERGY EFFICIENCY INCENTIVES DO THE COMPANIES OFFER?

A. Our understanding is that the Companies offer prescriptive, custom, and direct install incentives as part of their energy efficiency programs. For prescriptive measures, incentives can range from $25 to $12,000 per qualifying measure installed. Custom incentives are paid on a per-therm basis, with a maximum cap of up to $100,000 in incentives. Finally, direct install incentives provide free installation of low-cost heating and water saving measures, but no direct payment is made to the customer. Instead, a vendor provides the direct install measure and is paid based on the quantifiable savings achieved thereby.

Q. ARE THESE INCENTIVES PAID UPFRONT?

A. In most cases, no. According to the Companies' response to City IR-122, prescriptive incentives are paid to a customer after the qualifying equipment already has been purchased, and after a detailed verification and documentation process. For custom incentives, customers must complete an application form, schedule a pre-inspection, and submit the required documentation (*e.g.*, cut-sheets, usage data, anticipated project costs, and other technical data). Once the technical review is complete, a Company commitment letter indicating an authorized incentive level is provided to the customer. Once the project is completed, and a post-inspection completed, the incentive is issued. With respect to direct install incentives, the customer must first contract with a qualifying vendor to perform the necessary energy efficiency work.

We do note that, with respect to prescriptive incentives provided via the Companies' online Market Place, qualifying products and quantities can be purchased at a discounted price. In other words, Market Place prescriptive incentives are provided at the point of sale.

Q. DO YOU SEE ANY ISSUES WITH THE INCENTIVE PAYMENT MECHANISM AS DESIGNED?

A. As stated above, the difficulty for some buildings owners to obtain project financing for their energy efficiency investments can prevent otherwise meritorious and beneficial projects from being undertaken, or prevent building owners from undertaking deep retrofits in favor of smaller, less-costly (and likely less beneficial overall) measures. To the extent that the Companies do not provide incentives or

other financial support to a building owner until an energy efficiency project is completed, it can exacerbate this issue.

Q. DO YOU HAVE ANY RECOMMENDATIONS FOR SOLUTIONS?

A. Yes. The Companies should provide funding at the outset for some projects. Project developers then would be able to rely on this funding to obtain financing for the projects, either using the funding as collateral to reduce the financing needed, or as a source of money to repay the financier. To be clear, we propose that the Companies provide a limited amount of funding for this purpose ($1-3 million), at first, to test and verify the concept, and that it be used in partnership with private, 3rd party financing. These efforts will allow for additional private capital to be leveraged in support of larger energy efficiency investments and more comprehensive retrofits.

Q. PLEASE DESCRIBE YOUR PROPOSAL.

A. We propose that KEDNY and KEDLI each segregate a portion of the budgets allotted to their respective multifamily energy efficiency program offerings. The Companies would then encourage building owners and project developers to apply by adjusting the project incentive award timeline to assist the owners in obtaining private financing. Otherwise, the projects would be evaluated and selected in the same manner as currently occurs. Upon selection, the applicable Company would provide a letter verifying the anticipated funding amount the building owner or project developer would receive under the standard program parameters. The verification letter will be contingent on a commitment by a third party financier.

The funding amount could be held in an escrow account by the Companies, the Companies' implementation contractor, or the participating 3rd party lender until the project is complete, then released to the appropriate party. Company funding could be provided subject to repayment if the project is not completed. Under this proposal, the building owner or project developer would be able to use the funds from the Companies to assist in obtaining financing for the project. This approach also should facilitate the deployment of private-market funds for energy efficiency projects and extend the positive impact of incentive dollars much further.

Q. WHAT IS THE PROPOSED TERM FOR THIS PROPOSAL?

A. We propose that it last for the earlier of exhaustion of the funding or three years. Upon its conclusion, the Companies should conduct an evaluation of the performance and default rates to ascertain whether the expected savings are achieved and what the risk of loss is. If this approach results in greater levels of energy efficiency project implementation and corresponding energy savings, and the risk of default, or losses, is minimal, then consideration should be given to broader relaxation of the requirement that funding is provided only after projects are completed.

Q. WHO SHOULD BE THE TARGET OF THIS PROPOSAL?

A. This proposal should target owners of multifamily housing, subsidized housing, limited equity cooperatives, and other low and moderate income residential buildings, who we believe are most likely to benefit from the revised funding structure. These owners often have less cash flow and more difficulty in obtaining

financing for large projects. Providing funding upfront could help them and their tenants obtain the benefits of deep energy efficiency savings that otherwise may be unattainable. The proposal should also target projects where incentive payments are expected to exceed a certain dollar threshold. We propose $50,000 as that threshold.

Q. SHOULD UPFRONT FUNDING BE AVAILABLE TO CUSTOMERS WHO OPT TO INSTALL CUSTOM EFFICIENCY MEASURES?

A. Yes. For customers who choose to install custom energy efficiency measures, the Companies should offer a commitment letter indicating the maximum authorized incentive level. The commitment letters can contain caveats that full payments are not provided until the projects are completed and verified, and that failure to satisfactorily complete the committed projects as proposed will result in the customer being required to reimburse the Companies for any upfront payments provided thereby. The commitment letters (including any conditions) should be designed such that financiers are comfortable with their terms. The combined provision of upfront funding and a commitment letter prior to commencement of projects for purposes of securing financing should increase the number of projects, help leverage private capital in support of energy efficiency, and help achieve the State's, City's, and the Companies' energy efficiency goals

Q. DO YOU HAVE ANY OTHER RECOMMENDATIONS ON ENERGY EFFICIENCY PROGRAM DESIGN?

A. Yes. The Companies should also tailor their energy efficiency program offerings to incentivize customers to pursue comprehensive packages of energy efficiency

measures that, together, could result in greater energy savings than discrete measures alone. For example, the Companies could offer higher incentives when related measures are coupled together, such as an increased incentive for new heating and cooling equipment provided a building owner also agrees to make improvements to the building envelope – air sealing, replacing windows and doors, adding insulation, etc.

Combining incentives associated with multiple improvements should encourage or facilitate more comprehensive efficiency measures and produce larger savings. For example, the magnitude of the incentives could increase non-linearly. That is, there is a greater overall value of performing five measures than one or two, so the incentives for multiple measures could be larger than simply adding together the incentives applicable to individual measures if performed discretely. Alternatively, a bonus incentive could be offered when a customer completes multiple measures as part of a comprehensive energy efficiency retrofit that achieves more than 20 percent energy savings.

Q. SHOULD THIS APPROACH FACTOR IN THE TIME IT TAKES TO COMPREHENSIVELY RETROFIT A BUILDING?

A. Yes. Given the expanded nature of this approach, the energy efficiency plan for a customer may need to extend over a period of years. The Companies should design their incentives to accommodate comprehensive upgrades that occur in stages, or over a longer period of time, rather than all at once. Instead of rushing through project scoping to meet prescribed deadlines, customers should be able to

implement projects that maximize savings. This adjustment also would improve coordination with private financing providers, allowing customers to leverage incentive funding to access additional capital to cover the up-front and related costs associated with building improvements not covered by the Companies' funding. Without this adjustment, customers are mostly reliant on self-financing, which further limits the scale and scope of efficiency projects. We recommend this change also be deployed generally.

The above recommendation notwithstanding, we note that we are not experts on the utilities' Technical Resource Manual, but it is our understanding that the Manual does not account for interactivity between measures. As a result, there is no motivation for program implementers or building owners to pursue a combination of comprehensive energy efficiency initiatives.

Q. WHAT ARE YOUR RECOMMENDATIONS REGARDING FUTURE ENERGY EFFICIENCY OFFERINGS?

A. There is precedent for creating a collaborative or other process to occur either as a later part of a rate case or during a rate year to explore issues that are better pursued through discussion than testimony and a hearing. This is such an issue. We recommend that the Commission convene either a technical conference or a collaborative process at which the details of the Companies' energy efficiency programs can be presented and discussed. This forum would allow interested stakeholders to offer and debate proposals for improvements to the Companies' current plans and programs, such as our inclining incentive proposal. Assuming

that consensus can be reached on one or more improvements, they can be presented to the Commission for approval and immediate implementation.

Q. DO YOU HAVE ANY OTHER RECOMMENDATIONS ON ENERGY EFFICIENCY?

A. Yes. The City believes it important that the use of the funds approved for the Companies' energy efficiency programs, as well as program performance and the savings actually achieved, be closely monitored. To that end, we recommend that the Companies be required to file a report no less frequently than twice per year containing the following information:

- Details on all expenditures made, broken down by program and including both the reporting period and periods starting with the inception of the program;
- Summary-level information on savings achieved, broken down by program and including both the reporting period and periods starting with the inception of the program; for programs that do not result in direct energy savings, information on the benefits realized from the program should be provided;
- For each program, detailed information on Dth, or MBtu avoided and GHG and peak load reductions; the savings should be broken down by building typology, geographic location – such as zip code, market sector, and technology(ies) deployed;

- Identification of constrained areas and the details we just described for each program that helps to alleviate the constraints;
- Identification of programs and/or specific projects targeted to low income customers, and the details we just described for each program or project, as applicable;
- A discussion of any program changes made over the reporting period and the reasons for such changes;
- A discussion of any program changes that will be made prospectively and the reasons for making the changes.

Additionally, once a year, the Companies should include in their report a discussion of lessons learned and best practices based on its prior experiences. This information would be helpful in designing future programs, and in modifying existing energy efficiency programs to maximize the benefits to customers.

IMPROVED COORDINATION WITH THE RETROFIT ACCELERATOR

Q. PLEASE DESCRIBE THE RETROFIT ACCELERATOR.

A. The City's Retrofit Accelerator has been a growing success and has gained broad acceptance from building owners, developers, contractors, and others. The Retrofit Accelerator was launched in September 2015 and offers a team of energy efficiency advisors to provide independent, customized, advisory services at no cost to building owners to help them complete cost-saving energy efficiency projects. The Retrofit Accelerator team assists building owners and decision-makers in navigating the building retrofit process in order to help participants reduce building

energy consumption and improve the comfort of tenants. This assistance involves identifying financing and incentive options, including those offered by the Companies, and provides guidance for participation in these programs.

Since its launch in 2015, the participants in the Retrofit Accelerator have completed energy efficiency and clean energy projects in thousands of buildings in New York City, with expanded program offerings projected to reach even more buildings throughout the City in the near future. The touchstone of this program is providing free, personalized advisory services to streamline the process of energy efficiency improvements, including helping building decision makers navigate available incentive programs.

Q. IS THERE ANY COLLABORATION BETWEEN THE COMPANIES AND THE RETROFIT ACCELERATOR?

A. Yes. The Retrofit Accelerator website currently offers an "NYC Incentive Map" that helps customers to find programs available in New York City to assist in building retrofits, including energy efficiency rebates and incentives offered by the Companies. For example, the NYC Incentive Map provides a high-level overview of the Companies' Multifamily Program, such as the types of measures covered by the program and the possible incentive amounts, and then provides a convenient link to the dedicated utility website for that program.

The Retrofit Accelerator also has collaborated with the Companies in the past with respect to its 2012 NYC Clean Heat initiative to address the public health hazard presented by heavy heating oil emissions in New York City. As part of the

Clean Heat Initiative, the City worked closely with utilities (including the Companies), contractors, financiers, and others to provide customers the ability to easily work through the process to convert building boilers from heavy fuel oil (*i.e.*, No. 6 fuel oil) to natural gas and No. 2 fuel oil, in order to comply with a regulation that required all use of No. 6 fuel oil to be phased out by 2015. Due in large part to the efforts of the City, the utilities, contractors, and financiers, the Clean Heat Initiative has been very successful.

Q. ARE THERE OPPORTUNITIES TO IMPROVE THE LEVEL OF COORDINATION BETWEEN THE COMPANIES AND THE RETROFIT ACCELERATOR?

A. We believe so. From a customer perspective, having one location in which to obtain information about available programs, apply for incentives, find contractors, and, as necessary, secure financing, would be ideal. In the case of energy efficiency, an approach similar to the coordinated approach taken with the Clean Heat Initiative would help to ensure that consumers are getting all information about available programs and funding.

Q. WHAT ARE YOUR RECOMMENDATIONS?

A. There should be greater coordination of messaging, marketing, and outreach between the Companies' energy efficiency programs and the Retrofit Accelerator program. Closer coordination and stronger synergies between the two initiatives will strengthen the positive perception of both programs in the public eye, reduce

duplication of benefits, and result in a clear and unified approach to cut greenhouse gas emissions and improve efficiency.

We also recommend a direct link between the Companies' energy efficiency programs and the Retrofit Accelerator program managers. When customers seek information or assistance via the Retrofit Accelerator, the personnel assisting the customers should be able to contact knowledgeable Company personnel directly and immediately regarding questions about available energy efficiency programs and incentives, and to seamlessly enroll customers in those programs.

We also recognize that the collaborative approach should not be one-sided. That is, the Companies also should work closely with Retrofit Accelerator personnel to identify incremental opportunities for energy efficiency beyond what is provided through the Companies' programs. The Retrofit Accelerator offers services for both new construction and existing buildings, as well as a High Performance Retrofit Track, that are utility and provider agnostic. These services can help drive down city-wide energy demand and increase efficiency for buildings that otherwise might not be eligible for utility programs, or when projects seek assistance completing projects outside the current scope of utility program offerings.

Another process improvement that can be made is closer coordination of customer enrollment. For example, the Retrofit Accelerator currently follows standard industry practice in completing a comprehensive intake survey of all new program participants. To the extent that a Retrofit Accelerator contact is then

transferred to the Companies, there should not be a need for the intake process to be repeated on the Companies' end. We recommend that the Companies' energy efficiency department work with Retrofit Accelerator staff to develop a shared, or common application that can be used by both programs. This will significantly reduce program administrative costs for both parties and best meet customer needs by alleviating the frustration expressed by our shared customers, who are currently being asked to duplicate the intake process.

Lastly, we recommend that the Companies track all customer referrals made to and from the Retrofit Accelerator, including what projects were pursued, and the project status. Consistent and transparent reporting will provide both the Companies and Retrofit Accelerator staff greater insight into where and how customers originate. It also will allow us to quantify the frequency of customer referrals and value of any customer acquisition the Retrofit Accelerator provides. We further recommend that direct customer referrals made by the Retrofit Accelerator to the Companies' programs not be counted as customer acquisitions made by the Companies, and that any associated dollar value be made available to support additional program related outcomes.

EXPANDING DEMAND RESPONSE EFFORTS

Q. WHAT IS THE CITY'S POLICY ON GAS DEMAND RESPONSE?

A. As explained in more detail by the City's Policy Panel, the City generally is supportive on enhanced energy efficiency and demand response efforts. Demand response programs can help to minimize reliance on natural gas by reducing the

need for gas capital expenditures to accommodate growing demand. In turn, such programs can help achieve both the City's and State's clean energy policy goals. In light of the continued uncertainty related to the NESE pipeline project, it is even more crucial now that the Companies have plans in place to effectively manage their current and future gas systems, including through the use of demand response to reduce peak load.

Q. ARE YOU FAMILIAR WITH THE COMPANIES' EXISTING GAS DEMAND RESPONSE DEMONSTRATION ("DRD") PROJECT?

A. Yes. We understand that the City has been an active participant in the DRD project since 2016, and that its experience has generally been positive. Given how well the program has worked to date, we believe that there is an opportunity here to significantly expand on the existing framework of the DRD Project to achieve even greater natural gas consumption and peak load reductions across the downstate National Grid service territories.

Q. HAVE YOU REVIEWED THE COMPANIES' PROPOSAL TO EXPAND THE DRD PROJECT?

A. Yes. Our understanding is that the Companies plan to expand the DRD Project's peak demand reduction targets from 80 to 1,000 Dth per hour of demand reduction for KEDNY, and from 60 to 400 Dth per hour of demand reduction for KEDLI, by 2024. Our understanding is based on the Companies' response to City IR-89 and the information set forth in Schedule 5 of Exhibit__(FOH-1). The Companies also are proposing to include a new penalty mechanism to address program

participants who fail to respond to called curtailment events, or who fail to meet minimum committed reductions.

Q. WHAT ARE YOUR OPINIONS ON THE PROPOSAL?

A. Generally, we support the Companies' decision to expand the DRD Project, particularly given how successful the program has been over the last few years. However, we are concerned that, given recent developments with the NESE pipeline, the proposal does not go far enough to address the lingering concern that growing gas demand will outstrip gas supply in New York City.

Q. DOES THE CITY HAVE ANY SPECIFIC RECOMMENDATIONS ON THE EXPANDED DRD PROJECT?

A. The City's recommendations on the DRD Project are set forth in the Direct Testimony of both the NYC Rate Design Panel and the NYC Policy Panel. Briefly, the Companies should increase the targets of the DRD Project by, among other things, expanding the pool of eligible customers, reducing the requirement for a 10 percent or more committed reduction, increasing customer recruitment, or through other measures. The Companies also should explore the expansion of the DRD Project through a collaborative that includes interested stakeholders

<u>CONCERNS REGARDING THE POWER-TO-GAS PROGRAM</u>

Q. HAVE YOU REVIEWED KEDLI'S PROPOSED P2G DEMONSTRATION PROJECT?

A. Yes.

Q. WHAT IS YOUR OPINION ON THE P2G DEMONSTRATION PROJECT?

A. The City conceptually supports the theory behind the P2G demonstration project, and supports the Companies in evaluating its feasibility. Converting excess renewable energy into pipeline quality natural gas can help to reduce the carbon footprint of the upstream gas system, and can allow the Companies to meet some of their gas system requirements with RNG. This is consistent with both the City's and State's clean energy goals.

We note that in order for P2G to play a meaningful role in decarbonizing the gas system, there first would have to be an excess of renewable energy to convert into methane through this process. The Companies should ensure that, if P2G becomes an integral part of the strategy to increase RNG use, it relies only on excess renewable electricity, and does not otherwise prevent customers from consuming that renewable electricity in the first place. P2G also should not be used as a substitute for comprehensive efforts to reduce overall emissions, gas usage, and peak demand, for example, through expanded energy efficiency efforts.

EXPANDING UTILITY FLEET ELECTRIFICATION

Q. PLEASE DESCRIBE THE CITY'S VEHICLE ELECTRIFICATION POLICY GOALS.

A. The City's goal in the transportation sector is to have an 80 percent sustainable mode share by 2050, meaning that four in five trips in New York City will be by foot, bicycle, or public transit. However, we recognize that sustainable modes may not be accessible for all New Yorkers, which is why for the trips that still need to be taken by vehicles, we want to target zero emission vehicles.

This transportation sector goal is part of a larger goal established by Mayor Bill de Blasio for New York City to reduce citywide greenhouse gas emissions by 80 percent by 2050, based on a 2005 baseline. Transportation accounts for approximately 30 percent of citywide greenhouse gas emissions. The bulk of on road transportation greenhouse gas emissions come from passenger vehicles. According to the *New York City's Roadmap to 80x50* prepared by the Mayor's Office of Sustainability in October 2016, a transition to low carbon fuels and vehicles could account for up to a 13 percent reduction in transportation emissions.

Q. WHAT TYPES OF VEHICLES ARE OF MOST INTEREST TO THE CITY WITH REGARDS TO ITS TRANSPORTATION POLICY?

A. Electric vehicles, or EVs, are of particular interest to the City since they produce no tailpipe emissions and are significantly more energy efficient than their internal combustion engine counterparts. However, the City also sees merit in vehicles that rely on alternatives to fossil fuels, such as renewable natural gas ("RNG").

Q. WHAT IS THE CURRENT COMPOSITION OF THE COMPANIES' FLEET, BY FUEL TYPE?

A. According to the Companies' response to City IR-99, the Companies currently operate a fleet of 275 vehicles that run solely on compressed natural gas ("CNG"), the vast majority of which are located in the Long Island and New York City regions. The Companies also appear to be operating a number of all-electric vehicles as part of their fleets, though most of these vehicles consist of forklifts and small construction vehicles such as backhoes. Importantly, according to their

response to City IR-97, the Companies' fleets currently only include a handful of all-electric or hybrid passenger vehicles.

Q. DOES NATIONAL GRID HAVE ANY PLANS OR INTERNAL POLICIES TO ELECTRIFY ITS PASSENGER VEHICLE FLEET?

A. According to National Grid US's website, and the Companies' response to City IR-98, National Grid has pledged to spend 5 percent or more of its annual acquisition dollars on plug-in EVs and technologies (including charging stations and infrastructure). This 5 percent commitment is cumulative across all of National Grid's businesses. Per the response to City IR-99, the Companies also indicated that National Grid has a general policy to accelerate the process of converting its fleet of passenger cars to all EV vehicles during 2019 and 2020.

Importantly, however, it does not appear that the Companies have any policy to convert their own fleet vehicles to alternative fuels outside of the general policy that broadly covers all National Grid businesses. In response to City IR-124, the Companies stated that they have concrete plans to convert a small number of (*i.e.*, 6) diesel and propane-powered forklifts to electric models, and to collaborate with an established manufacturer on a trial of an all-electric towable backhoe in 2019. With respect to passenger vehicles, the Companies only stated that they expect to continue converting such vehicles to EV, but there are no finalized plans for 2020.

Q. ARE THE COMPANIES' PLANS CONSISTENT WITH THE CITY'S POLICIES?

A. No. In our opinion, the Companies need to do more to accelerate the conversion of their existing fleets of passenger cars to EVs. At present, it does not appear that the Companies have any concrete plans in the near-future to significantly accelerate the conversion of fleet passenger cars to EVs.

Q. WHY IS IT IMPORTANT FOR THE COMPANIES TO ACTIVELY SEEK TO CONVERT THEIR FLEETS TO ALTERNATIVE FUELS?

A. As noted above, in order to achieve an 80 percent reduction in citywide greenhouse gas emissions by 2050, significant in-roads will have to be made with respect to reducing the emissions impact of the transportation sector. This includes electrifying the significant fleet operations that operate in New York City, including the fleets operated by the Companies.

Electrifying the Companies' fleets can produce positive public health benefits. For example, air quality in New York City has been an ongoing concern for many years. Data from the New York City Department of Health and Mental Hygiene shows that areas with large populations of low income residents have 1.7 times the exposure to annual average of fine particulates and over nine times the rate of hospital visits for asthma due to diesel emissions from mobile sources than more affluent areas. Similarly, both gasoline and diesel vehicles contribute to ozone formation that, when inhaled, can have detrimental health impacts such as increasing the likelihood of asthma attacks. By converting their existing gasoline- and diesel-fired vehicles to all-electric, or even to lower emissions fuel types such

as CNG, the Companies can significantly reduce the air quality and health impacts of their vehicles' emissions on the City's population.

Q. WHAT DO YOU RECOMMEND?

A. The Companies should commit to increased EV adoption for their fleets beyond the existing 5 percent goal spread across all National Grid businesses. Specifically, for new light duty fleet purchases, the City recommends that each Company commit at least 25 percent of its annual acquisition budget toward EVs, with the goal of increasing that commitment to 50 percent within five years.

Electric vehicles have a potentially lower overall cost of ownership than their internal combustion engine counterparts (mostly through maintenance and fueling cost reduction). The City has found that its electric fleet vehicles have a maintenance cost of between $204 and $386 annually. Conventional gasoline fleet vehicles, however, cost the City between $922 and $1,805 annually.

In addition, to the extent that acquiring EVs for the Companies' fleets is not feasible (for example, for heavy duty vehicles), the Companies should nonetheless commit to transitioning those vehicles toward CNG- or RNG-fueled vehicles. If the Companies decide to focus on acquiring CNG vehicles in the near term, it should develop plans to transition those vehicles to RNG within 10 years.

EXCESSIVE EARNINGS ADJUSTMENT MECHANISMS

Q. WHAT ARE EARNINGS ADJUSTMENT MECHANISMS ("EAMS")?

A. In our understanding, earnings adjustment mechanisms, or EAMs, are positive-only shareholder incentives designed to align utility incentives with policy goals,

primarily the objectives of Reforming the Energy Vision, or REV, and carbon reduction.

Q. WHAT IS YOUR OPINION OF EAMS?

A. In general, the City is not opposed to providing a utility with shareholder incentives if it performs at an exceptional level. However, incentives need to be designed solely or primarily to reward exceptional performance, not standard or expected behavior, and the amount must be measured and reasonable to avoid unduly burdening customers. As an example, in an Order issued on April 24, 2019 in Cases 16-M-0429 and 14-M-0101, related to interconnection EAMs, the Commission specifically indicated that an EAM should be "an effective measure of exceptional utility performance." We also believe that EAMs should be tied to some demonstrable action taken by the utility, and that utility shareholders should not be rewarded for actions taken solely by third parties, or for circumstances over which the utilities have no control.

We also note that the fact that an EAM, inclusive of its shareholder incentives, may have a cost-benefit ratio in excess of 1.0 is not sufficient to demonstrate exceptional performance or warrant payment of shareholder incentives. The size of the incentives, their impact on customers, and the quality of the performance leading to the outcome must be considered.

Q. HAVE YOU REVIEWED NATIONAL GRID'S EAM PROPOSALS?

A. Yes.

Q. WHAT EAMS DO THE COMPANIES PROPOSE?

A. The Companies propose three EAMs: (i) System Efficiency, split into Peak Reduction and Supply Diversification metrics; (ii) Energy Efficiency, split into Incremental Energy Efficiency and Affordability metrics; and (iii) Carbon Reduction, split into a Full Service metric and a Transportation metric.

Q. DO YOU HAVE ANY GENERAL CONCERNS ABOUT THE PROPOSED EAMS?

A. Yes. We are concerned that, as currently proposed, National Grid's EAMs are excessive in terms of absolute dollar values. Moreover, the City does not believe that actions taken by third parties should form the basis for shareholder incentives, or that utilities should be rewarded when it is unclear if they are taking any actions to achieve specific policy-oriented results. We also are concerned that some of the proposals may be duplicative, and that the Companies potentially could be rewarded multiple times (through different EAMs) for a single occurrence or achievement.

Q. PLEASE EXPLAIN WHY YOU BELIEVE THE COMPANIES' CURRENT EAM PROPOSALS ARE EXCESSIVE.

A. According to the Companies' responses to City IRs 65 and 66, KEDNY and KEDLI could potentially earn at least $25.75 million and $16.42 million, respectively, in 2020 if they achieve the maximum possible award for each of their proposed EAMs. These incentive levels jump to $30.04 million for KEDNY and $19.55 million for KEDLI by 2023. We say "at least" because these amounts do not appear to include any potential earnings associated with the Companies' proposed

Platform Service Revenues, the Non-Pipes Alternatives incentive mechanism, or any positive revenue adjustments proposed by the Companies' Gas Safety Panel, which we do not address in this testimony.

To put this request into perspective, KEDNY could potentially reward its shareholders an amount that is larger than the proposed rate allowance for the utility's low income program during the Rate Year, and KEDLI could reward its shareholders an amount that is triple the Rate Year rate allowance for its low income program. We believe that the Companies' low income customers are more in need of additional benefits than their shareholders.

Q. DO YOU HAVE ANY CONCERNS ABOUT SPECIFIC EAMS?

A. Yes. We have concerns about the proposed Affordability metric, the Carbon Reduction metrics, and the Peak Reduction metric.

Q. PLEASE EXPLAIN YOUR CONCERNS WITH THE AFFORDABILITY METRIC.

A. Under the Affordability metric of the Energy Efficiency EAM, the Companies could earn up to ten 10 basis points for a two percent reduction in the average low-income gas heating customer's bill (or one basis point for every 0.2 percent reduction). In their response to DPS IR-262, the Companies reported that in calendar year 2018, the average bill for a low income heating customer was $125.63 in the KEDNY service territory, and $115.21 in the KEDLI service territory. Thus, for achieving a two percent reduction in those bills ($2.51 and $2.30, respectively),

KEDNY would earn $3.3 million in shareholder incentives and KEDLI would earn $2.1 million in shareholder incentives.

Moreover, according to their response to DPS IR-262, the Companies define "low-income" for the purposes of the Affordability metric as those gas heating customers enrolled in the Energy Affordability Program ("EAP"). In response to DPS IRs 260 and 261, KEDNY reported an average of 62,696 heating customers enrolled in its EAP during calendar year 2018, and KEDLI reported an average of 10,864 heating customers enrolled in its EAP.

Based on the foregoing 2018 figures, as proposed, assuming the Companies achieve the maximum possible savings under this metric (i.e., KEDNY's and KEDLI's EAP heating customers save an average of $2.51 and $2.30 per month, respectively), KEDNY's shareholders would receive $1.75 in incentives for every dollar that an average low-income customer saves during the year, while KEDLI's shareholders would receive $7.00 for every dollar that an average low-income customer saves during the year. These amounts were calculated by multiplying the monthly two percent reduction by 12 to get annual bill savings for an average EAP heating customer, then multiplying annual bill savings by total EAP heating customers to get total estimated annual savings for all low income heating customers, and then dividing total estimated annual savings into the maximum possible shareholder incentive. This disparity in shareholder rewards as compared to the corresponding benefits to low income customers is highly inequitable, and

demonstrates that National Grid's proposed Affordability metric needs substantial reworking.

Q. DOES THE AFFORDABILITY METRIC MEASURE ACTUAL PERFORMANCE BY THE COMPANIES?

A. The answer to this is unclear. The Future of Heat Panel's direct testimony does not specifically elaborate on the precise actions that the Companies intend to take to reduce low-income bills. Rather, the Panel explained on page 106 of its pre-filed Direct Testimony that this metric "incentivizes the Companies for programmatic efforts to reduce low-income customers' energy usage and for animating the market for third parties to likewise provide low-income benefits within the Companies' service territories." As this metric is calculated before any low income discounts are factored in (according to the Companies' response to DPS IR-262), achievement of the metric's targeted bill reduction appear to be independent of the Companies' existing low income programs.

The Future of Heat Panel also did not state whether the Affordability metric would account for factors aside from weather, so it is unclear to us if the Companies' shareholders would be allowed to earn an incentive if average low income customer bills are reduced by, for example, decreases in the cost of natural gas, reductions in usage from ratepayer-funded energy efficiency programs, or actions of third parties unrelated to the Companies' efforts. We note that the Companies have set the minimum baseline for this metric at one basis point for a 0.2 percent reduction of average low income customer bills, or a roughly 25 cent

reduction in average bills, which conceivably could occur as a result of factors outside of the Companies' control.

Q. WHAT IS YOUR RECOMMENDATION?

A. We recommend that the Affordability metric be rejected at this time. As described above, we have numerous concerns with this proposal, including: (i) the size of the shareholder incentive is disproportionate to the benefits derived by low income customers; (ii) the metric can be achieved without any effort by National Grid (*i.e.*, through gas cost reductions or third-party actions); (iii) there is a potential double-count if savings are achieved through ratepayer funded energy efficiency efforts (for which KEDNY and KEDLI are also seeking an EAM); and (iv) it is uncertain how performance will be measured under this metric.

EAMs should be designed to incentive exceptional utility behaviors and outcomes. With the Affordability metric, the Companies could potentially take no action and still receive a windfall, which is borne by customers. This is an unreasonable outcome.

Q. DOES THE CITY SUPPORT OTHER MEASURES TO ASSIST LOW INCOME CUSTOMERS?

A. Yes, absolutely. The City is committed to reducing the energy burden faced by its low income residents. That is precisely why the City's Low Income Panel is proposing to increase the size of the current low income discounts to amounts that are designed to satisfy the Commission's policy of achieving an energy cost burden of no more than six percent of a customer's income. We understand that

the City's Low Income Panel is making several other recommendations to improve the Companies' low income programs, including more frequent file matches with the NYC Human Resources Administration and elimination of reconnection fees for low-income KEDNY customers.

We contend that resources for low income customers are better directed to these efforts, rather than to shareholder incentives. In fact, for KEDNY and KEDLI, there is a combined $5.4 million in proposed shareholder incentives for the Affordability metric. These dollars would be better utilized as part of direct funding increases for existing low income programs.

Q. DO YOU HAVE CONCERNS WITH THE CARBON REDUCTION METRIC?

A. Yes. As proposed, the Carbon Reduction metric is composed of two metrics: (i) the Full-Service CO_{2e} reduction metric; and (ii) the Transportation CO_{2e} reduction metric. We have concerns with both metrics, which we address in turn.

Q. PLEASE EXPLAIN YOUR CONCERN WITH THE FULL-SERVICE CO_{2e} REDUCTION METRICS.

A. We are struggling to differentiate among the Supply Diversification EAM and the Full-Service CO_{2e} Reduction EAM, which appear as if they both could be triggered by the same outcomes. According to the Companies' response to City IR-95, the Supply Diversification EAM measures "the volume of RNG injected into the system," whereas the Full-Service CO_{2e} Reduction EAM "measures the amount of RNG purchased by customers of the Companies." In other words, the Companies

potentially could receive a shareholder incentive for the total supply of RNG that is injected into the gas system, and then again for the amount of that supply purchased by customers through a proposed Green Gas Tariff. This proposal is not reasonable and should be corrected.

Q. SHOULD THE UTILITY BE REWARDED TWICE FOR THE SAME ACHIEVEMENTS?

A. No. EAMs should be designed to reward utilities for achieving exceptional, discrete outcomes, but not in a manner where a single measure (for example, an energy efficiency measure) triggers multiple incentives for the Companies. Accordingly, we recommend rejecting the Full-Service CO_{2e} Reduction EAM.

Q. PLEASE EXPLAIN YOUR CONCERN WITH THE TRANSPORTATION CO_{2E} REDUCTION METRIC.

A. The metric would reward shareholders for achieving certain CO_{2e} reduction targets associated with switching vehicles from diesel or gasoline to CNG/RNG. As discussed herein, we are proposing that the Companies significantly increase their investment in vehicles that run either on electricity or RNG. These investments, if adopted, would result in overlap between ongoing, ratepayer funded efforts and the shareholder incentives proposed under the Transportation CO_{2e} reduction metric.

This proposal also raises a concern about potential double-counting with the Supply Diversification EAM, which measures "the volume of RNG injected into the system." The Future of Heat Panel notes on page 112 of its pre-filed Direct Testimony that, to the extent a fueling "station signs up to use RNG for a portion

of its load, the Companies will also incorporate those savings into the CO_{2e} calculation." In other words, the Companies are proposing to incentivize shareholders for both the injection of RNG into its system and the use of that RNG by a vehicle fueling station.

Q. WHAT DO YOU RECOMMEND?

A. Based on the concerns identified above, we recommend rejecting the full Carbon Reduction metric as proposed. This is not to say, however, that we are opposed to incentives that promote incremental RNG and geothermal deployments by the Companies. While we do not have a specific metric to offer in place of the Carbon Metric reduction, at a minimum any such metric should: (1) reward the utility for superior performance, not business-as-usual behavior; (2) require that the outcomes are driven by utility action, as opposed to actions by third parties; (3) avoid double-counting with other incentives or metrics; and (4) be readily measurable and verifiable.

Q. PLEASE EXPLAIN YOUR CONCERNS WITH THE PEAK REDUCTION METRIC.

A. In the Peak Reduction EAM, the Companies set the baseline for earning an incentive at the forecasted peak reduction under their DRD Project. The Companies already are asking for a combined $2 million for the Project, per Schedule 5 of Exhibit__(FOH-1). If the project achieves its intended purposes – purposes for which National Grid proposes that customers pay for – it would be

entitled to an additional combined $543,000. This amounts to rewarding National Grid for what it is being paid to do.

Q. WHY IS IT PROBLEMATIC TO SET THE BASELINE FOR EARNING AN INCENTIVE AT THE FORECASTED PEAK REDUCTION UNDER THE DRD PROJECT?

A. As we noted above, EAMs should be designed to reward superior performance by a utility, not business-as-usual performance. However, by using the proposed minimum targets of program to establish the minimum threshold for the Companies to earn an EAM on that program, the Companies essentially are rewarding themselves for doing what they would have done anyway.

Q. DO YOU HAVE ANY OTHER COMMENTS ABOUT THE COMPANIES' PROPOSED EAMS?

A. Yes. We note that not all of the proposed EAMs and metrics appear to be rewarding the Companies for business-as-usual performance, or for occurrences out of their control. For example, as explained in the Future of Heat Panel's direct testimony, the Companies are proposing shareholder incentives for incremental MMBtu reductions achieved through their expanded energy efficiency programs, as directed by the Commission in the Energy Efficiency Order. Importantly, the Companies set the minimum threshold for their respective Incremental Energy Efficiency metrics equal to the targets set in the Energy Efficiency Order, and assigned them basis points earnings opportunities of *zero.* In other words, the Companies have appropriately identified the Energy Efficiency Order targets as

baseline targets, and only are proposing to reward themselves for achieving energy efficiency savings above those targets. The Companies should endeavor to design all of their EAMs in such a manner.

Despite our agreement with the *design* of shareholder incentives for the Energy Efficiency EAM, we do not agree with the actual target levels proposed by the Companies for their Energy Efficiency EAM, as discussed in more detail below.

Q. WHAT DO THE COMPANIES PROPOSE FOR THEIR ENERGY EFFICIENCY REDUCTION METRIC?

A. Per their responses to DPS IR-387 and City IR-92, the Companies propose a minimum threshold for their Energy Efficiency metric based on targets set forth in the Energy Efficiency Order, including incremental savings targets. The minimum target is then assigned zero basis points. The maximum target was developed by assuming cost efficiencies equal to each Company's respective lowest average historical run rate from the last four years (adjusted for anomalies and to remove behavioral program results). The mid-point target for the metric is set in between these two points.

Q. DO YOU AGREE WITH THE PROPOSED ENERGY EFFICIENCY METRIC TARGETS?

A. No. As noted above, over the last several years, the unit costs for energy efficiency savings have declined. For example, KEDNY spent an average of $1.64 per therm of energy efficiency savings in 2018, while KEDLI spent an average of $1.61 over

the same period. The Companies should be lauded for improving the effectiveness of their respective energy efficiency programs over time. However, by the same token, new shareholder incentives should encourage the Companies to build upon these successes, not to reward them for continuing business-as-usual. Here, the Companies could each potentially earn 40 basis points for achieving cost efficiencies that they already have achieved in recent years.

Q. HOW SHOULD THE ENERGY EFFICIENCY METRIC TARGETS BE SET?

A. We recommend that the Companies set their minimum targets by assuming cost efficiencies equal to each Company's average historical actual run rate over the last four years. In other words, the current proposed maximum targets should instead be set as the minimum targets. Midpoint and maximum targets should then be set at 10% and 20% savings above the minimum, respectively. This appropriately incentivizes the Companies to strive for superior performance, as opposed to rewarding them for business-as-usual performance.

Q. DOES THIS CONCLUDE YOUR TESTIMONY?

A. Yes.